Shrishgovind Umesh Revankar

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EDUCATION

Stevens Institute of Technology | Hoboken, NJ, United States **Expected May 2026**
Master of Science, Computer Engineering | **GPA: 3.5** | **Concentration:** Embedded Systems
Relevant Courses: Real-Time and Embedded Systems, Digital & Computer Systems Architecture, Applied Data Structure & Algorithms, Introduction to VLSI Design, Internet of Things, Autonomous Mobile Robotic System.
Fr. Conceicao Rodrigues College of Engineering | Mumbai, Maharashtra, India **May 2022**
Bachelor of Engineering, Electronics Engineering

SKILLS

Programming & Software Development: Embedded C, C++, Python, .NET Framework.
Embedded Systems & Firmware: Bare-Metal Programming, .NET Desktop Application.
Communication Protocols: CAN (Controller Area Network), SPI, I2C, UART, Wi-Fi, Bluetooth.
Tools & Development Environment: Proteus, Keil uVision5, LTSpice, Vitis & Vivado, Logic Analyzer, MATLAB.
Hardware Skills: Wire & PCB Soldering, Signal Debugging, PCB Designing, Waveform Generator, EMI/EMC Testing.

WORK EXPERIENCE

Tork Motors Pvt. Ltd. | Pune, India **June 2022 – May 2024**
Research & Development Engineer

- Architected and deployed a Hardware-In-Loop (HIL) system that improved production efficiency by 20% of the motorcycle's Vehicle Control Unit and added data encryption to improve the traceability of the VCU.
- Developed production-cluster firmware for 3-Wheeler, to be configurable to multiple vehicle variants, achieving standardization of the code base and delivering a 10% improvement increase in overall system efficiency.
- Initiated the development of a proprietary software application using .NET Framework for automating CAN tests, streamlining the testing processes, improving the testing process by 30%, and with additional test cases.

Kalyani Powertrain Pvt. Ltd. | Pune, India **January 2022 – February 2022**
Intern

- Researched EV battery thermal management, focusing on thermal runaway prevention, and system safety.

CRCE Formula Racing – Formula Student (Formula Bharat) | Mumbai, India **March 2019 – September 2021**
Electronics Head

- Led the team to develop a CAN-based telemetry system integrating internal and external sensors for real-time vehicle monitoring, enabling performance tuning and design optimization.
- Engineered semi-automatic transmission system reducing chassis width and improving lap times by 20 seconds.
- Implemented a safety-critical Brake System Plausibility Device (BSPD) using analog comparators and timer ICs to detect brake-throttle conflicts, trigger a 10-second shutdown, and automatically restore vehicle operation

PROJECTS

Smart Environmental Monitoring System (Computing Principles for Mobile & Embedded Systems) **May 2025**

- Developed bare-metal firmware for STM32U5 (B-U585I-IOT02A) interfacing with onboard temperature and humidity sensors using I2C and SPI protocols.
- Acquired real-time environmental data and transmitted it to cloud platform via Wi-Fi module.

Embedded Security System (Computing Principles for Mobile & Embedded Systems) **May 2025**

- Built bare-metal firmware on STM32 Nucleo for a proximity-based security system using ultrasonic sensing for intrusion detection.
- Integrated matrix keypad authentication with GPIO multiplexing and an OLED display for user feedback.

Space Shooter Game (Logical Design of Digital Systems) **Nov 2025**

- Designed and implemented a video game on a Xilinx Zynq FPGA using hardware-software co-design, generating real-time graphics output over HDMI, and input handling and game logic using onboard buttons.

Self-Balancing Robot Control System (Graduate Research Project) **May 2026**

- Developed a MATLAB-based control framework for a nonlinear two-wheeled self-balancing robot using state-space modeling, LMI-based stabilization, and data-driven control.
- Evaluated disturbance rejection and closed-loop stability through simulation of body angle, position, velocity, and control input responses.